Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the test accuracy during training.

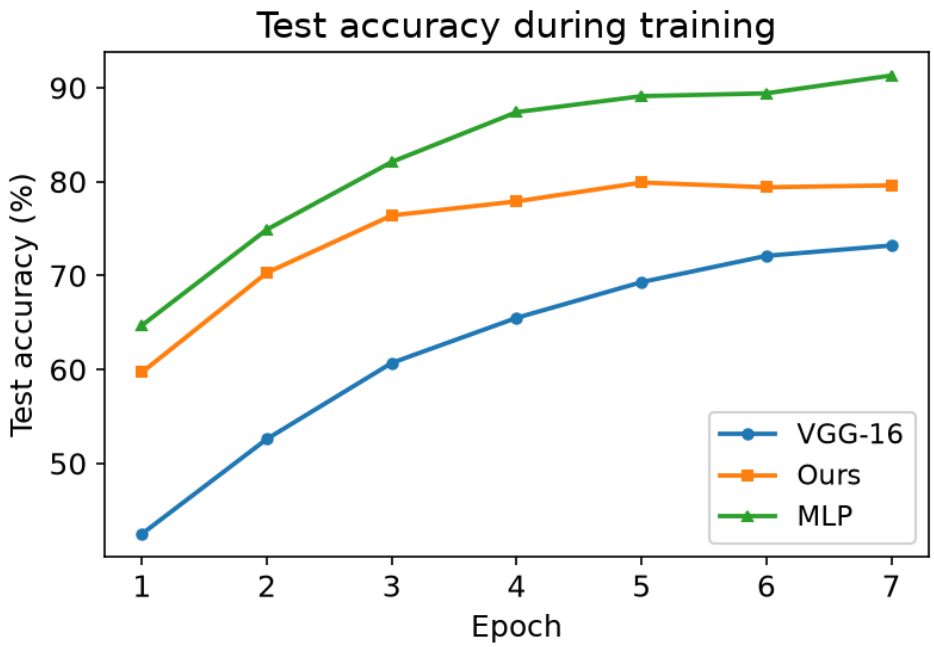


Figure 1: Test accuracy during training for the models in this experiment.

MLP attains a test accuracy of over 85% at epoch 4 (Figure 1). MLP outperforms VGG-16 by about 22 percentage points at epoch 2.

Figure 2 shows the test accuracy during training.

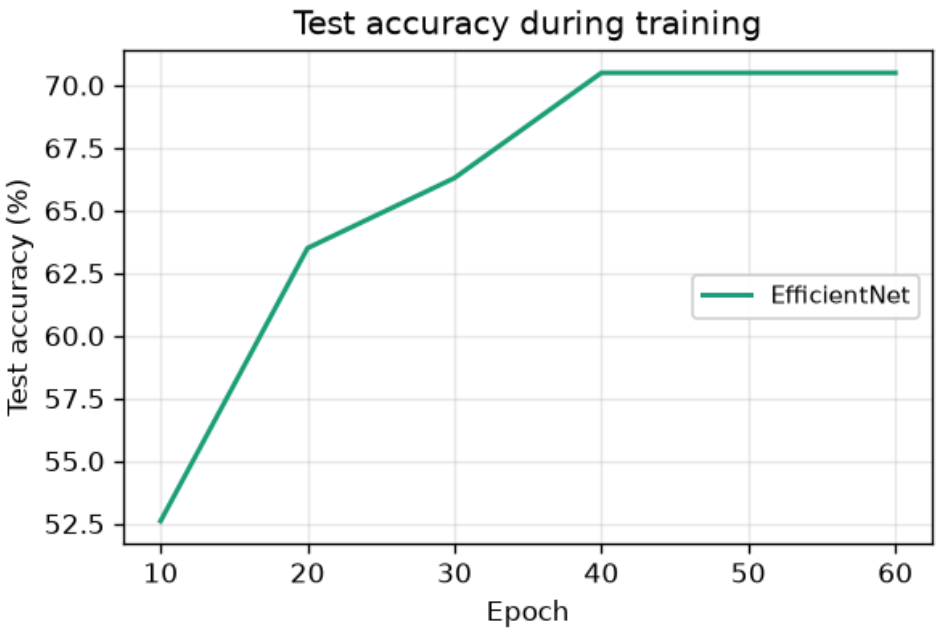


Figure 2: Test accuracy during training for the models in this experiment.

At epoch 20, EfficientNet achieves about 60% test accuracy. At epoch 60, EfficientNet achieves 70% test accuracy (Figure 2). EfficientNet scores 4.2 points higher on 40 than on 30 (Figure 2).

3 Discussion

SVM reaches 90.5% accuracy on AG News. We note that the training was stable for all models.